

A MULTIPHYSICS APPROACH TO LFR ANALYSIS

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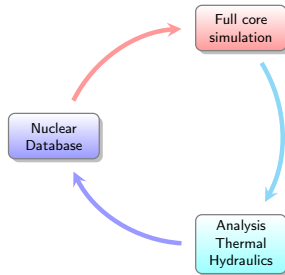
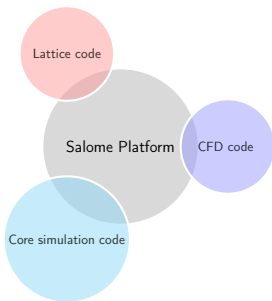
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Fast Reactors of the SMR Type, 26 September 2019

Introduction of multiphysics approach

Development of computational platform that integrates different computational tools, into the common wrapping framework given by the **SALOME platform**:

- Lattice code, **DRAGON**
- Full core simulation code, **DONJON**
- 3D-porous thermal-hydraulic CFD code, **FEMus**



Targets of multiphysics approach

- Investigate the coupling between neutronic code and thermal-hydraulic code
- Implementation of multiphysics approach into preliminary model of a Fast Small Modular Reactor (SMR)
- Choosing a lead cooled reactor conceptual design, ALFRED, characterized by small core dimension along the axis

Future perspectives

A coupled neutronics and thermal hydraulics analysis to better supporting the fast core dynamics (in both fields) at design level

Neutronic Code

The neutronic code is an [open-source reactor physics code](#) "[Dragon-Donjon Ver5.0.5](#)" developed at Ecole Polytechnique de Montreal (Canada), designed around methods to solving transport equation of neutron.

Lattice code, DRAGON

Simulate the neutronic behavior of the fuel assembly in the reactor core and create the proper nuclear database, called MULTICOMPO

Full core simulation, DONJON

Interpolate information contained in MULTICOMPO, simulate the comprehensive neutronic behavior of core and compute power and flux distribution

Nuclear Database - MULTICOMPO

MULTICOMPO

- **data structure** which contain nuclear properties of assembly's materials.
- possibility to recovery data concerned parameters as **nuclide's temperature and density or depletion calculation**.
- necessity to create an optimal nuclear database for a correctly use of full core simulation code.

- ① Isotopic cross-section library JEFF 3.1.2 with 315 energy groups
- ② Lattice calculation
 - Burn-up calculation
 - Temperature and density modification
 - Homogenization and collapse to 33 energy groups
- ③ Saving macroscopic cross sections in MULTICOMPO

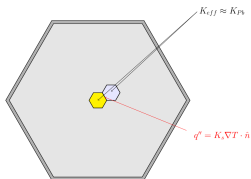
Computational fluid dynamics code

FEMus

is an open-source version, developed at the University of Bologna, of the LIBMESH library with parallel MPI-PETSC and multigrid solvers

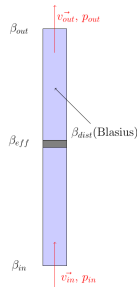
The CFD solvers

have the function to solve energy and temperature equations starting from a given neutronic power density distribution



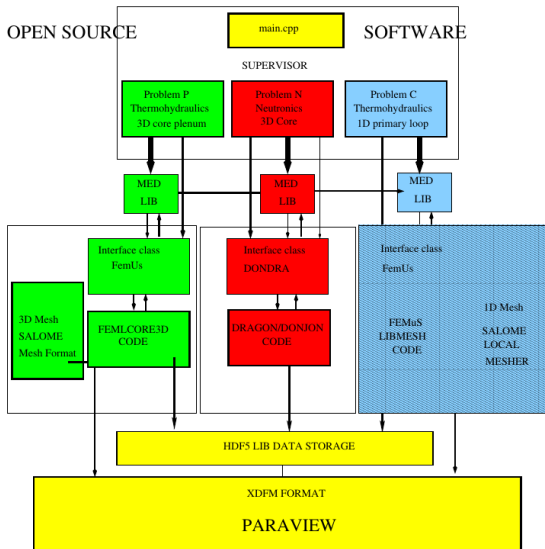
Assumption for solution of energy equation is porous medium

$$\vec{v} \cdot \nabla T + \nabla K_{eff} \nabla T = Q_s$$



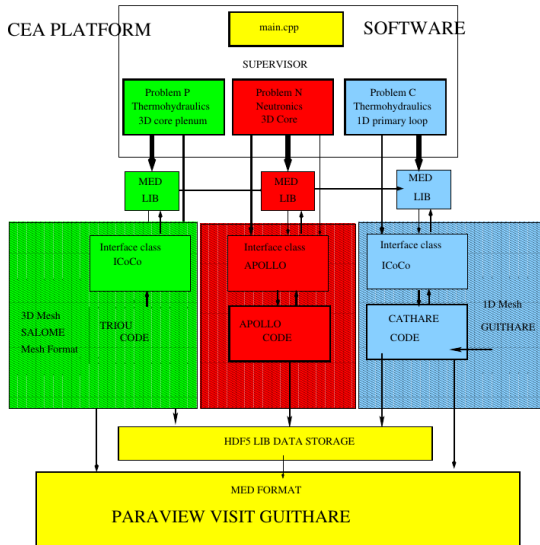
Computational platform

Open source codes



Computational platform

Proprietary codes



Computational wrapping platform

The main purpose

is to couple the solution of a multi-physics and multi-scale three-dimensional problem inside a simplified and more comprehensive framework.

The Salomé project

facilitates the coupling of scientific mesh-based codes thanks to its architecture and suite of tools that provide several data interfaces and exchange across the different codes

The integration of a code on the Salomé platform is obtained by generating an interface with functions available in the MEDMem library that **allows a data transfer from the platform to the code and then from the code to the interface**

Main modules used in wrapping

MED module

provide a library for **storing** and **recovering** computer data in a suitable format

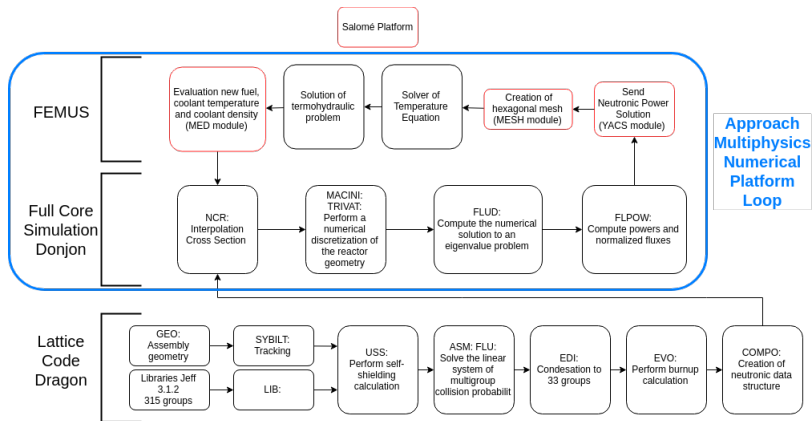
GEOM and MESH modules

have the main function to **draw** and **create meshes** in MED format, respectively.

YACS module

is a tool to **supervise execution** of complex interconnected scientific applications as object structure available during execution of DONJON code.

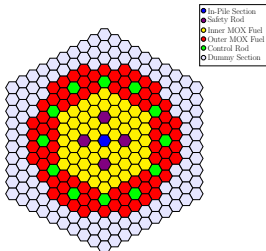
Summary flowchart



The ALFRED (SMR) concept model

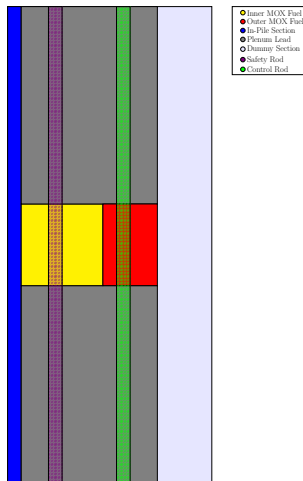
Data and Modeling

Parameters	Unit	Value
Thermal Power	MW	300
Total height vessel	mm	3500
Inner vessel inner/outer radius	mm	1475/1525
Lattice pitch (hexagonal)	mm	13.86
Wrapper thickness	mm	4.0
Distance between to wrappers	mm	5.0
Pins per FA	-	126
Total FA	-	134
Inner/Outer FA	-	56/78
Inner/Outer enrichment	%	20.5/26.2
Control/Safety Rods	-	12/4
In-pile section	-	1
Average Core Flow	m/s	1.28
Coolant Inlet/Outlet Temperature	°C	400/520



Reactor core fuel map

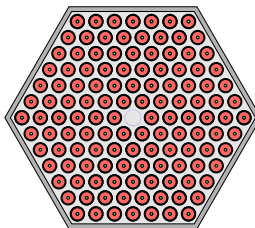
Sketch of the reactor core



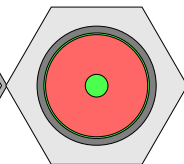
Fuel pin and assembly

Data and Modeling

Parameters	Unit	Value
Pellet hollow diameter	mm	2.0
Pellet radius	mm	4.5
Gap thickness	mm	0.15
Clad thickness	mm	0.6
Pin diameter	mm	10.5
Bottom plug length	mm	50
Gas plenum height	mm	550
Bottom plug length	mm	10
Active height	mm	600
Upper insulator height	mm	10
Spring length	mm	120
Upper plug length	mm	50



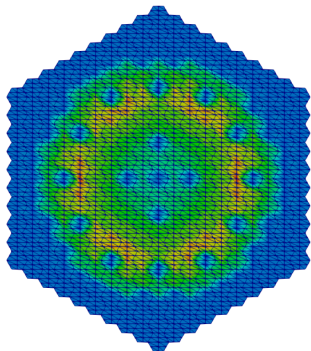
Fuel Assembly



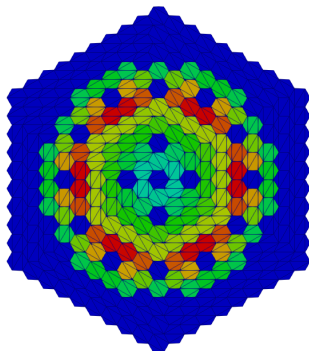
Fuel pin

Finite elements for neutronic and CFD codes

Mesh CFD

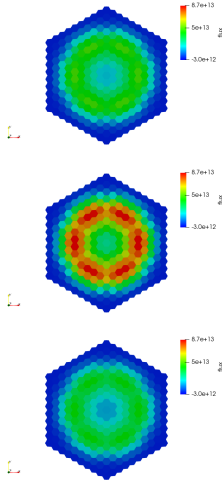


Mesh Neutronic

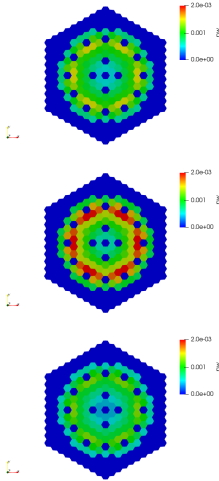


Preliminary result of multiphysics approach

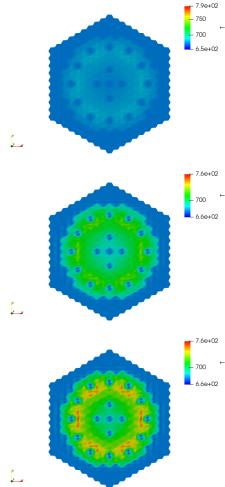
Flux distribution



Power distribution

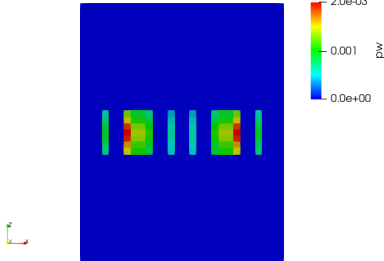


Temperature field

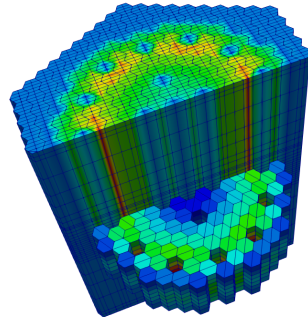


Preliminary result of multiphysics approach

Power distribution along axial



Wapping power distribution with
temperature field on reactor
section



Conclusions

